

the excluded periods $\ell' = -1$ or 1 .

The example shown in Figure 1 provides a visualization of three takeaways regarding the contamination in the pre-trend coefficient μ_{-2} . First, both panels show that weights are a non-linear function of cohort shares. Second, panel (a) confirms the structure for weights associated with $\ell' \neq \ell$ but $\ell' \in g^{incl}$ as described in Proposition 1, namely $\sum_e \omega_{e,\ell'}^{-2} = 0$ for $\ell' \neq -2$. However, these weights have non-zero magnitude. When the effect is homogenous across cohorts, the contaminations are equal to $\sum_e \omega_{e,\ell'}^{-2} ATT_{\ell'}$ and cancel each other out. In contrast, when effects are heterogeneous the different $CATT_{e,\ell'}$ will not necessarily cancel and will contaminate the estimate for μ_{-2} . Third, panel (b) confirms the structure for weights associated with excluded periods as described in Proposition 1, namely $\sum_e \omega_{e,\ell' \in g^{excl}}^{-2} = -1$. In other words, contaminations from excluded periods can be thought of as a type of “normalization”: a weighted average of excluded $CATT_{e,\ell'}$ is subtracted off the estimated treatment effect. However because these weights are not contained in $[0, -1]$, this average may lie outside of the convex hull of $CATT_{e,\ell'}$ for excluded periods. The latter issue can be alleviated by an assumption that $CATT_{e,\ell'}$ are the same in all excluded periods; we can fully avoid contamination from excluded period treatment effects by assuming all associated $CATT_{e,\ell'}$ are equal to zero.

4 Alternative estimation method

We propose a new estimation method that is robust to treatment effects heterogeneity. The goal of our method is to estimate a weighted average of $CATT_{e,\ell}$ for $\ell \in g$ with reasonable weights, namely weights that sum to one and are non-negative. In particular, we focus on the following weighted average of $CATT_{e,\ell}$, where the weights are shares of cohorts that experience at least ℓ periods relative to treatment, normalized by the size of g :

$$\nu_g = \frac{1}{|g|} \sum_{\ell \in g} \sum_e CATT_{e,\ell} Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}. \quad (25)$$

One can aggregate $CATT_{e,\ell}$ to form other parameters of interest, such as those proposed by Callaway and Sant’Anna (2020a). We focus on the above aggregation ν_g since our goal is to improve the non-convex and non-zero weighting in μ_g . The weights in ν_g are guaranteed to be convex and have an interpretation as the representative shares corresponding to each $CATT_{e,\ell}$. Thus, our alternative estimator $\widehat{\nu}_g$ improves upon the two-way fixed effects estimator $\widehat{\mu}_g$ by estimating an interpretable weighted average of $CATT_{e,\ell \in g}$.

Our method proceeds by replacing each component in ν_g with its consistent estimator. We first estimate

each $CATT_{e,\ell}$ using an interacted two-way fixed effects regression, then estimate the weight $Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ using their sample analogs. In the final step, we average over the cohort-specific estimates associated with relative period ℓ . This method has a similar flavor as the method proposed by Gibbons et al. (2019). They first use an interacted model to estimate the treatment effect for each fixed effect group; the resulting group-specific estimates are averaged to provide the ATE. Their method improves fixed effects regressions in a cross-sectional setting, and our method builds on theirs by improving two-way fixed effects regressions in a panel setting. We therefore follow their terminology in calling our alternative estimator an “interaction-weighted” estimator.

4.1 Interaction-weighted estimator

We describe the estimation procedure in three steps (with more detailed definitions stated in Definition 4 of Online Appendix B).

Step 1. We estimate $CATT_{e,\ell}$ using a linear two-way fixed effects specification that interacts relative period indicators with cohort indicators, excluding indicators for cohorts from some set C :

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{e \notin C} \sum_{\ell \neq -1} \delta_{e,\ell} (\mathbf{1}\{E_i = e\} \cdot D_{i,t}^\ell) + \epsilon_{i,t}. \quad (26)$$

The exact specification depends on the cohort shares for a given application. If there is a never-treated cohort, i.e. $\infty \in \text{supp}\{E_i\}$, then we may set $C = \{\infty\}$ and estimate regression (26) on all observations. If there are no never-treated units, i.e. $\infty \notin \text{supp}\{E_i\}$, then we may set $C = \{\max\{E_i\}\}$, i.e. the latest-treated cohort and estimate regression (26) on observations from $t = 0, \dots, \max\{E_i\} - 1$. Lastly, if there is a cohort that is always treated, i.e. $0 \in \text{supp}\{E_i\}$, then we need to exclude this cohort from estimation.

The coefficient estimator $\widehat{\delta}_{e,\ell}$ from regression (26) is a DID estimator for $CATT_{e,\ell}$ with particular choices of pre-periods and control cohorts. As DID is likely a familiar estimator for applied researchers, we separate the more in-depth discussion of its definition, choices of pre-periods, and choices of control cohorts in Section 4.2.

Step 2. We estimate the weights $Pr\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ by sample shares of each cohort in the relevant period(s) $\ell \in g$.

Step 3. To form our IW estimator, we take a weighted average of estimates for $CATT_{e,\ell}$ from Step 1

with weight estimates from step 2. More formally, the IW estimator is

$$\widehat{v}_g = \frac{1}{|g|} \sum_{\ell \in g} \sum_e \widehat{\delta}_{e,\ell} \widehat{Pr}\{E_i = e \mid E_i \in [-\ell, T - \ell]\} \quad (27)$$

where $\widehat{\delta}_{e,\ell}$ is returned from step 1 and $\widehat{Pr}\{E_i = e \mid E_i \in [-\ell, T - \ell]\}$ is the estimated weight returned from step 2. We normalize the weights further by the size of g . If g is a singleton, then its size is $|g| = 1$.

Validity of the IW estimator. Under the parallel trends and no anticipation assumptions the coefficient estimator $\widehat{\delta}_{e,\ell}$ from regression (26) is a consistent estimator for $CATT_{e,\ell}$. The sample shares of each cohort are also consistent estimators for the population shares. Thus, the IW estimator is consistent for a weighted average of $CATT_{e,\ell}$ with weights equal to the share of each cohort in the relevant period(s).

With a few standard assumptions (which we present as Assumption 4 in Online Appendix B) on regression (26), we can show that each IW estimator is asymptotically normal and derive its asymptotic variance. The large sample approximation allows us to estimate the variance of IW estimators directly without relying on bootstrapping as in Callaway and Sant'Anna (2020a). However, we only construct pointwise confidence interval valid for a given IW estimator $\widehat{v}_g$. The bootstrap-based inference by Callaway and Sant'Anna (2020a) constructs simultaneous confidence intervals that are valid for the entire path of $\widehat{v}_g$.

4.2 Difference-in-differences estimator for $CATT_{e,\ell}$

Definition 2. Assume cohort e is non-empty i.e. $\sum_{i=1}^N \mathbf{1}\{E_i = e\} > 0$. Assume there exists some pre-period $s < e$ and some set of control cohorts $C \subseteq \{c : e + \ell < c \leq T\}$ that are non-empty i.e. $\sum_{i=1}^N \mathbf{1}\{E_i \in C\} > 0$. Using the notion $\mathbb{E}_N$ to abbreviate the symbol $\frac{1}{N} \sum_{i=1}^N$, the DID estimator with pre-period s and control cohorts C estimates $CATT_{e,\ell}$ as

$$\widehat{\delta}_{e,\ell} = \frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]} - \frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i \in C\}]}{\mathbb{E}_N[\mathbf{1}\{E_i \in C\}]} \quad (28)$$

The assumption of non-empty cohort e , existence of pre-period and non-empty control cohorts makes the DID estimator well-defined. For example, DID estimators for cohort 0 are not well-defined because a pre-period does not exist for this cohort, which is why we exclude them from estimating regression (26) in Step 1.

Using the above definition, in regression (26) from Step 1 of our proposed method, the coefficient

estimator $\widehat{\delta}_{e,\ell}$ is a DID estimator for $CATT_{e,\ell}$ with pre-period $s = e - 1$ (because we exclude relative period $\ell = -1$) and some choice of control cohorts C . If there is a never-treated cohort i.e. $\infty \in \text{supp}\{E_i\}$, then we set the control cohort to be never-treated units $C = \{\infty\}$. If there are no never-treated units, i.e. $\infty \notin \text{supp}\{E_i\}$, then we set the control cohort $C = \{\max\{E_i\}\}$, i.e. the latest-treated cohort. Among all possible pre-periods, for regression (26) from step 1 of our proposed method we choose $s = e - 1$ and $C = \{\infty\}$ with never-treated units (or $\{\max\{E_i\}\}$ without never-treated units) because the resulting specification is a natural extension of the common specifications of two-way fixed effects regression.

Note that without never-treated units we need to drop time periods $t \geq \max\{E_i\}$ from estimating regression (26) because every unit will be treated in these periods. The DID estimators for $CATT_{e,\ell}$ for $e + \ell \geq \max\{E_i\}$ are thus not well-defined as the control cohort is empty $C = \emptyset$. For example, when there are just two cohorts with $E_i \in \{1, T\}$, one treated at $t = 1$ and the other treated in the last period, we then need to omit interaction terms involving the latest-treated cohorts as well as dropping observations from the last period $t = T$ from estimation.

Under some assumptions, this DID estimator $\widehat{\delta}_{e,\ell}$ is an unbiased and consistent estimator for $CATT_{e,\ell}$, a fact that we build on in deriving the probability limit of the IW estimator. We state this in the following proposition.

Proposition 5. *If Assumptions 1 and 2 hold, then the DID estimator using any pre-period $s < e$ and non-empty control cohorts C is an unbiased and consistent estimator for $CATT_{e,\ell}$.*

It is possible to relax the parallel trends assumption to allow the timing of treatment to depend on covariates. One can estimate $CATT_{e,\ell}$ consistently based on the inverse propensity score reweighted estimator proposed by Abadie (2005) and Callaway and Sant’Anna (2020a), the outcome regression approach by Heckman et al. (1997), and the doubly robust estimator recently proposed by Sant’Anna and Zhao (2020). The resulting estimates $\widehat{\delta}_{e,\ell}$ can then be plugged into step 3 to form our IW estimator. In particular, without covariates and for the case with never treated units, our approach coincides with Callaway and Sant’Anna (2020a). Therefore one can use the did R package developed by Callaway and Sant’Anna (2020b) to form the IW estimator.

The choice of the pre-period s and the control cohorts C depends on the trade-off between relaxing Assumption 1 or Assumption 2. If Assumption 2 is likely to hold, we can choose any pre-period $s < e$ and include not-yet-treated units in control cohorts $C = \{c : c > e + \ell\}$ as in Callaway and Sant’Anna (2020a). This